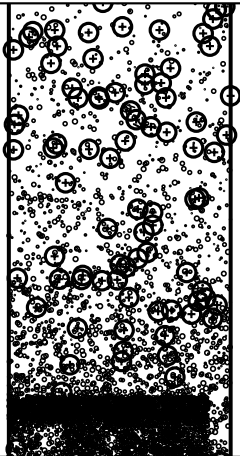


Iron Shot Dual-Role Agitation Mechanism (Real Gas Drag Only, Physical Lid, 1.5 mm Iron @ 3.5 m/s)

FIG. 3

Good-var checkpoint (real drag, physical lid):
Iron mean height: 34.47 mm
Regolith mean height: 11.56 mm
EMI: 3.58× vs no-iron baseline
Containment: 100.0% inside lid
N=6500, 100% physical-lid data only



Iron shot (1.5 mm, dual-role)



Regolith fines (mobilized by collisions)

Static Sintered Distributor (gas injection)